

Team 16 Happy Cows Release Notes

Link to our repo: <https://github.com/ucsb-cs156-s26/proj-happycows-s26-16.git>

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What's New (User Highlights)

Announcement Management

Admins now have full control over site-wide and commons-specific announcements. This includes a new functional Create Page, an Edit Page, and a management table. Players can now see announcements directly on their Play Page, ensuring they stay informed about game updates and events.

Happy Cows

Admin ▾

Welcome, [iholiday@ucsb.edu](#)

Log Out

Create Announcement for Commons Happy Farms

Start Date

dd/mm/yyyy --:--

End Date

dd/mm/yyyy --:--

Announcement

Create

Cancel

HappyCows is a project of [Mattanjah de Vries](#), Distinguished Professor of Chemistry at UC Santa Barbara.

Announcements for Commons: Happy Farms

Create Announcement

id	Start Date	End Date	Edit	Delete	Announcement
1	06/12/2026, 5:19 PM	06/14/2026, 7:21 PM	Edit	Delete	Make sure to submit your participation checks!
2	06/02/2026, 9:27 PM	06/08/2026, 11:59 PM	Edit	Delete	There are too many cows! Your cows will start losing health at this point.

Edit Announcement for Commons Happy Farms

Id

1

Start Date

12/06/2026 17:19

End Date

14/06/2026 19:21

Announcement

Make sure to submit your participation checks!

Update

Cancel

Howdy farmer Jack!

Announcements

test announcement

Today is day2!

Total Players: 1

Leaderboard

Full Course Management Interface

The administrative interface for managing courses is now complete. Admins can easily add new courses through a dedicated form and view all existing courses in a standardized, easy-to-read table format.

Backend Controller:

Course		^
GET	/api/course/{id}	Get a course by id
PUT	/api/course/{id}	Update a single course
DELETE	/api/course/{id}	Delete a single course
POST	/api/course	Create a course
GET	/api/course/all	List all courses

Index Page:

Courses

id	Code	Name	Term	Edit	Delete
2	CS156	Computer Science 156	s26	Edit	Delete

Create Page:

Create Course

Course Code (e.g. 'CHEM 123' or 'INT 94MW')

CS156

Name

Computer Science 156

Term

s26

Create Cancel

Edit Page:

Edit Course

Id

2

Course Code (e.g. 'CHEM 123' or 'INT 94MW')

CS178

Name

Computer Science 178

Term

s26

Update

Cancel

Commons Search Functionality

Finding a specific commons is now much faster. A new search box has been integrated into the Commons list page, allowing users to filter through available commons in real-time by listing them by name. This expands search capabilities beyond just sorting by categories.

Commons



f

Download All Stats

Happy Farm (ID: 2)		Hidden: false	
Cow Price: 100	Milk Price: 1	Start Balance: 10000	Starting Date: 2026-05-26
Last Date: 2026-06-26	Degrad Rate: 0.001	Show Leaderboard: false	Show Chat: false
Total Cows: 0	Cap / User: 50	Carry Cap: 100	Eff Cap: 100
Edit Delete Leaderboard Stats CSV Announcements Chat Dashboard			

My Farm (ID: 4)		Hidden: false	
Cow Price: 100	Milk Price: 1	Start Balance: 10000	Starting Date: 2026-05-26
Last Date: 2026-06-26	Degrad Rate: 0.001	Show Leaderboard: false	Show Chat: false
Total Cows:	Cap / User:	Carry Cap:	Eff Cap:

New Admin Dashboard

We've introduced a centralized Admin Dashboard. This new page serves as a starting point for various administrative tasks and can be accessed quickly via the new "Dashboard" button added to the admin navigation.

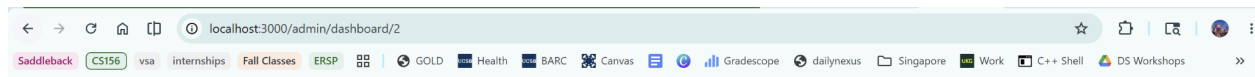
Admin Dashboard Button from the ListCommonsV2

Commons

[Download All Stats](#)

test (ID: 1)		Hidden: false	
Cow Price: 100	Milk Price: 1	Start Balance: 10000	Starting Date: 2026-05-13
Last Date: 2026-06-13	Degrad Rate: 0.001	Show Leaderboard: false	Show Chat: false
Total Cows: 0	Cap / User: 50	Carry Cap: 100	Eff Cap: 100

Admin Dashboard Page (Only Stubs)



Dashboard

Commons ID: 2

Overview

Total Farmers	Total Cows	Commons Balance	Days Active
--	--	--	--

Commons Details

Name: --
Status: --
Start Date: --

Participation Metrics

Active Farmers: --
Inactive Farmers: --
Avg Cows per Farmer: --

Farmer Cow Distribution

Histogram / distribution of cows per farmer will go here

Trends Over Time

Cows Over Time Chart placeholder (x-axis: days since commons start)	Farmers Over Time Chart placeholder (x-axis: days since commons start)
--	---

Additional Insights



Future analytics and breakdowns will be added here.

Chat Moderation Tools

To help maintain a positive community environment, admins now have the ability to monitor chats within a commons. This feature includes a user-friendly interface to view message history and the power to delete specific messages when necessary. Your messages are coded in blue, and everyone else's in grey.

Admin Chat Button from the ListCommonsV2

test (ID: 1)		Hidden: false	
Cow Price: 100	Milk Price: 1	Start Balance: 10000	Starting Date: 2026-05-13
Last Date: 2026-06-13	Degrad Rate: 0.001	Show Leaderboard: false	Show Chat: false
Total Cows: 0	Cap / User: 50	Carry Cap: 100	Eff Cap: 100
Edit Delete Leaderboard Stats CSV Announcements Chat			

Admin Chat Page

The screenshot shows the Admin Chat Page for Commons #1. The page header includes the HappyCows logo, navigation links (H2Console, Swagger, Admin), a welcome message for nikitahuynh@ucsb.edu, and a Log Out button. The main content area displays a list of chat messages. The first message is a deleted message from Nikita Huynh. The subsequent two messages are from Nikita Huynh with test2 and test1 as content. Each message has a Delete button. The page footer includes a logo and text stating that HappyCows is a project of Mattanjah de Vries, Distinguished Professor of Chemistry at UC Santa Barbara.

localhost:8080/admin/chat/1

HappyCows H2Console Swagger Admin Welcome, nikitahuynh@ucsb.edu Log Out

Back

Commons #1 • Admin Read Only Commons #1

🚫 = Deleted Message

Nikita Huynh
hiiiiiiiiii 2026-05-21 17:19:26

🚫 Message Deleted

Nikita Huynh
test2 2026-05-21 17:19:23 Delete

Nikita Huynh
test1 2026-05-21 17:19:21 Delete

[no more messages]

HappyCows is a project of Mattanjah de Vries, Distinguished Professor of Chemistry at UC Santa Barbara.

Ivy Holiday

Hey John, what's up ????

2026-05-21 18:09:38

Message Deleted

John Sylvester

qawertfhuokl

2026-05-21 18:09:22

Message Deleted

Manual Job Launching: Record Common Stats

Admins no longer need to wait for scheduled tasks to see updated statistics. A new launcher has been added to the Admin Jobs page, allowing for the manual execution of the "Record Common Stats" job.

Record Common Stats

Record statistics for all commons. This will create a CommonStats record for each commons with current health and profit data.

Record Stats

id	Created	Updated	Status	Log
19	5/14/2026, 17:31:26	5/14/2026, 17:31:27	complete	Starting record common stats job... Starting Commons id=1 (meow farm)... CommonStats 9 for commons id=1 (meow farm) finished. Record common stats job done!

UI Polish: Improved "Day" Display

We've corrected a formatting issue where "Day" and the number of days were missing a space, resulting in a cleaner and more professional-looking interface.

Ahoy! Welcome farmer Ivy!

Announcements

Today is day 1!

Total Players: 1

Leaderboard

Developer Notes

Backend Improvements

- **Security & Exceptions:** Introduced `HiddenCommonsException` to prevent unauthorized or accidental API executions on commons that are marked as hidden.

```
"id": 1,  
"name": "Test",  
"cowPrice": 100,  
"milkPrice": 1,  
"startingBalance": 10000,  
"startingDate": "2026-05-16T00:00:00",  
"lastDate": "2026-06-16T07:00:00",  
"showLeaderboard": false,  
"showChat": false,  
"capacityPerUser": 50,  
"carryingCapacity": 100,  
"degradationRate": 0.001,  
"hidden": true,  
"belowCapacityHealthUpdateStrategy": "Linear",
```

Error: response status is 400

Response body

```
{  
  "message": "Commons with id 1 is hidden",  
  "type": "CommonsHiddenException"  
}
```

- **Feature Flag Support:** Added `featureFlags` to the `SystemInfo` endpoint. This allows the application to dynamically enable or disable features without requiring a full redeploy.

Response body

```
{  
  "springH2ConsoleEnabled": true,  
  "showSwaggerUILink": true,  
  "startQtrYYYYQ": "20221",  
  "endQtrYYYYQ": "20243",  
  "sourceRepo": "https://github.com/ucsb-cs156/proj-happycows",  
  "commitMessage": "Merge pull request #72 from ucsb-cs156-s26/Jack-CommonsFeaturesEnumClean",  
  "commitId": "eb81faf",  
  "githubUrl": "https://github.com/ucsb-cs156/proj-happycows/commit/eb81faf",  
  "oauthLogin": "/oauth2/authorization/google",  
  "featureFlags": ""  
}
```

- **Enum Refactoring:** Cleaned up the `CommonsFeatures` enum by removing redundant values, leading to a more maintainable codebase.

Testing Improvements

- **Build Optimization:** Updated the `pom.xml` to skip certain dependencies and build steps when no changes are detected. This significantly reduces the time required for integration tests and local development cycles.

- Average time pre implementation: 3:30-4:15 per run of tests

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 13.11 s - in edu.ucsb.cs100.happiercows.integration
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 02:20 min
[INFO] Finished at: 2026-06-03T21:32:18-07:00
[INFO] -----
```

- Average time post implementation: 1:15-2:25 per run of tests

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 10.01 s - in edu.ucsb.cs100.happiercows.integration
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:29 min !!!
[INFO] Finished at: 2026-06-03T21:37:38-07:00
[INFO] -----
```

- **Dependency Updates:** Upgraded `jsdom` to version 16.7.0 to address testing environment issues and ensure compatibility with modern React testing patterns.